Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 2

Date: 21-04-2026

Smart Campus Event Management System

SDG: SDG No – SDG Title(SDG 4 – Quality Education)

Presented By:

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INTRODUCTION

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- In today's digital era, traditional methods of managing campus events lack efficiency, accessibility, and real-time coordination.
- Students often face difficulty in discovering, registering, and keeping track of upcoming workshops, seminars, and technical events.
- Administrators encounter challenges in organizing event details, managing registrations, and collecting feedback without a centralized system. This project addresses the need for a structured and efficient campus event management platform.
- The system allows users to register, log in, browse events, register for events, and submit feedback seamlessly.
- It ensures secure authentication using session management and password encryption through Spring Security.
- The application integrates Java Spring Boot and MySQL (XAMPP) to manage users, events, registrations, and feedback efficiently.

PROBLEM STATEMENT

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- Many educational institutions still rely on manual or semi-digital methods to manage campus events and student registrations.
- This leads to inefficiencies in organizing event details, tracking participant data, and managing schedules.
- Manual systems increase the risk of errors, data duplication, and loss of important information.
- Administrators face challenges in monitoring registrations and collecting feedback in real time.
- Students often miss important events due to lack of a centralized and accessible platform.
- There is a growing need for an automated and centralized system to manage campus events efficiently.
- Institutions require a reliable database system to manage users, events, registrations, and feedback.
- Hence, a web-based Smart Campus Event Management System is proposed to address these challenges.
- The system aims to improve accuracy, accessibility, security, and overall event management efficiency.

OBJECTIVE(S)

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- ✓ To design and develop a secure web-based Smart Campus Event Management System using Java Spring Boot and MySQL for efficient handling of users, events, registrations, and feedback.
- ✓ To implement user authentication and session management using Spring Security to ensure secure login, registration, and data protection.
- ✓ To provide an interactive and user-friendly interface that allows students to browse events, register easily, and submit feedback seamlessly.
- ✓ To enable administrators to create, update, delete, and manage events efficiently through a centralized dashboard.
- ✓ To automate event registration and maintain accurate records using a structured database system.
- ✓ To provide real-time access to event details, registration status, and feedback for better decision-making.
- ✓ To reduce manual work and improve operational efficiency through a centralized digital platform.
- ✓ To build a scalable system that can be extended with additional features such as notifications, analytics, and mobile access.

PROPOSED SYSTEM

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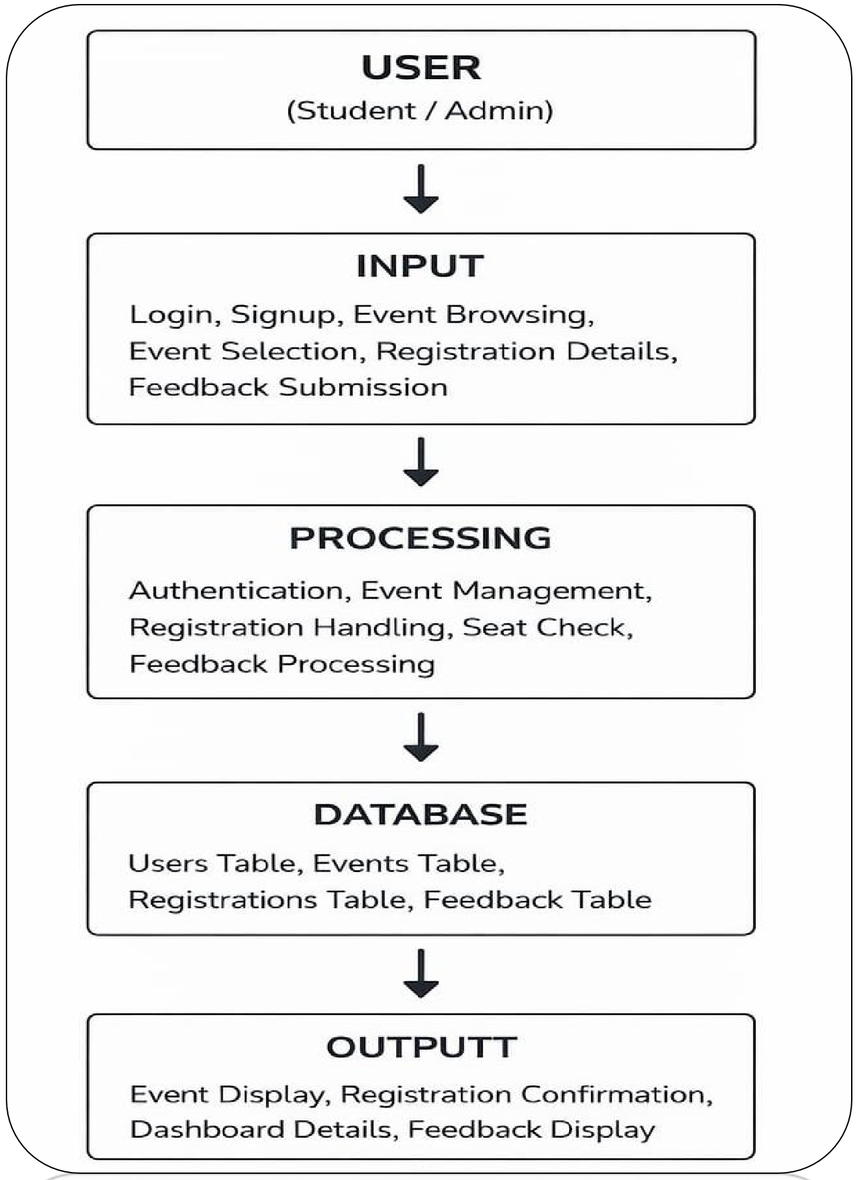
- The proposed solution is a web-based Smart Campus Event Management System developed using Java Spring Boot and MySQL.
- The system provides secure user registration and login using Spring Security with session management and password encryption.
- It allows students to browse events, register easily, and submit feedback through a user-friendly interface.
- Administrators can create, update, and delete events, and manage registrations through a centralized dashboard.
- All data related to users, events, registrations, and feedback are stored efficiently using a structured database.
- The system eliminates manual record-keeping issues and improves accuracy, speed, and reliability.
- Overall, it provides a scalable, secure, and efficient platform for managing campus events effectively.

PROPOSED SYSTEM ARCHITECTURE



Input:

Users enter registration details, login credentials, browse events, select events, and submit feedback through the web interface.

Processing:

The system verifies user authentication, manages event registrations, checks seat availability, and processes feedback using Spring Boot and session management.

Database:

All data such as users, events, registrations, and feedback are securely stored and managed in the MySQL database.

Output:

The system dynamically displays event listings, registration confirmations, feedback details, and user dashboard information on the web pages.

PROPOSED SYSTEM ARCHITECTURE

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SMART CAMPUS EVENT MANAGEMENT SYSTEM

HTTPS / (Fetch API)

CLIENT (Frontend)

- *HTML / CSS / JavaScript*
- *Thymeleaf Templates*
- *Login Page, Event Dashboard, Registration Page*
- *Form Validation & User Interaction*

HTTPS / (Fetch (JSON))

SERVER (Backend)

- *Java Spring Boot*
- *Spring MVC (Controllers)*
- *Spring Security (Authentication)*
- *REST APIs for Event & Registration Management*

SQL Queries

DATABASE

- *MySQL (XAMPP)*
- *Users*
- *Events*
- *Registrations*
- *Feedback*

3 – TIER ARCHITECTURE

❖ *Presentation Layer (Client)*

❖ *Application Layer (Server)*

❖ *Data Layer (Database)*

WORKFLOW

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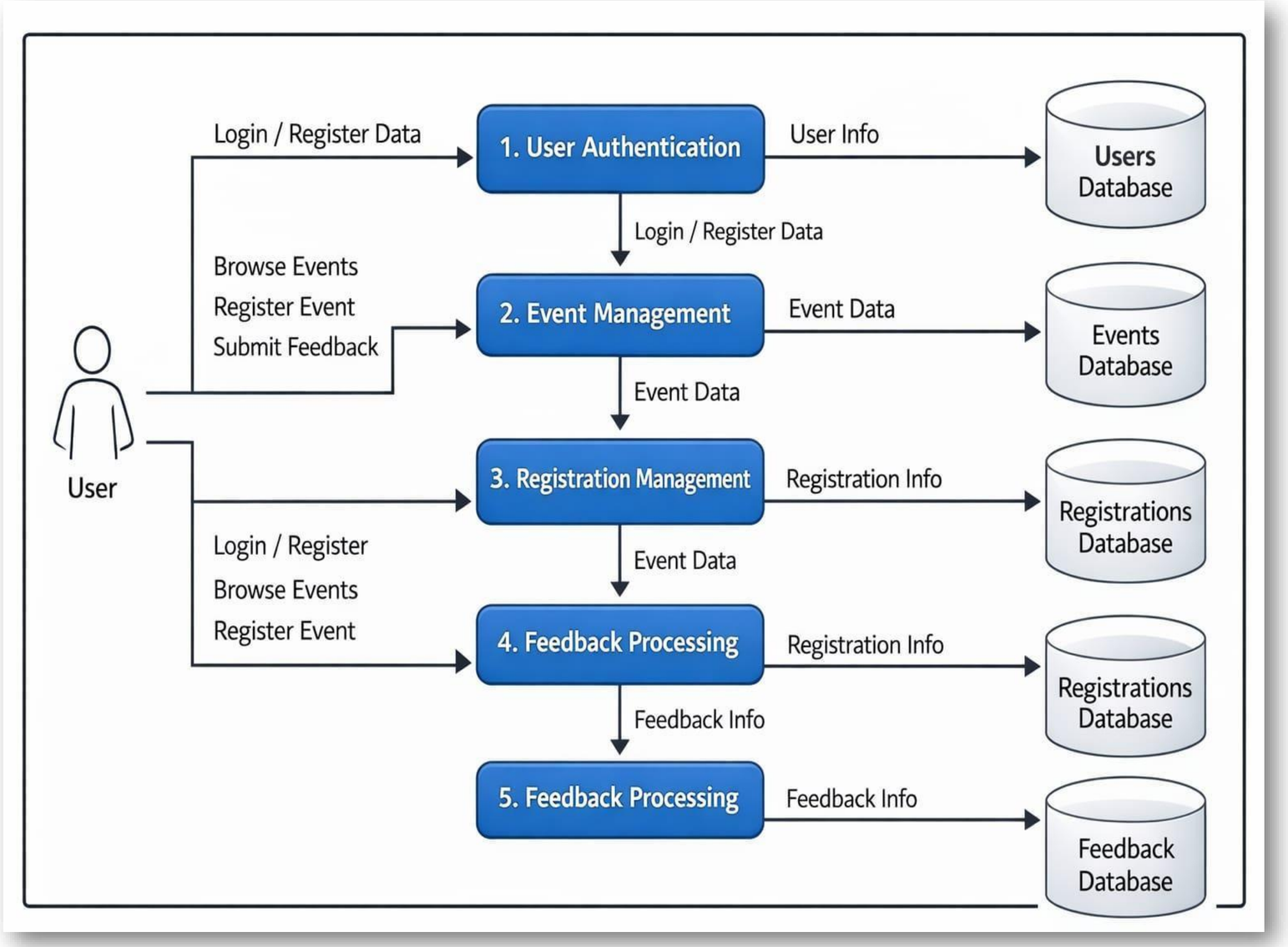
- The User (Student/Admin) logs in or registers and browses available campus events.
- The system retrieves event details from the MySQL Database (via XAMPP server).
- The user selects an event and completes event registration or submits feedback.
- The system processes the request and stores registration and feedback details in the database.
- The system generates outputs such as registration confirmation, event updates, and dashboard details.



TECHNOLOGIES USED

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💻 Frontend:

💻 **HTML5** – Used to structure web pages
🎨 **CSS3** – Used for styling and layout design
⚡ **JavaScript** – Used for validation and user interaction
🧩 **Thymeleaf** – Connects frontend with backend

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🗄️ Backend:

☕ **Spring Boot** – Handles server-side logic and APIs
🔄 **Spring MVC** – Manages request and response flow
🔐 **Spring Security** – Provides secure login authentication

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🗄️ Database:

🗄️ **MySQL (XAMPP)** – Stores user, event, and registration data

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🛠 Tools:

🔧 **XAMPP** – Used as a local server environment integrating Apache and MySQL to run the project.
💻 **Visual Studio Code** – Used as the code editor for development.
🌍 **Web Browser (Chrome/Edge)** – Used for testing and running the application.

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📜 Software Requirements:

- ☐ Windows 10/11 Operating System
- 🔧 XAMPP Server
- 🌐 Web Browser
- 📝 Code Editor (Visual Studio Code)
- ☕ Java JDK

MODULE DESCRIPTION

🧩 Module 1: User Management

- Handles user registration and login with secure authentication using Spring Security and session management.
- Ensures that only authorized users (Student/Admin) can access the system.

📰 Module 2: Event Management

- Displays available campus events with details such as date, venue, and description.
- Allows users to browse and view event information easily.

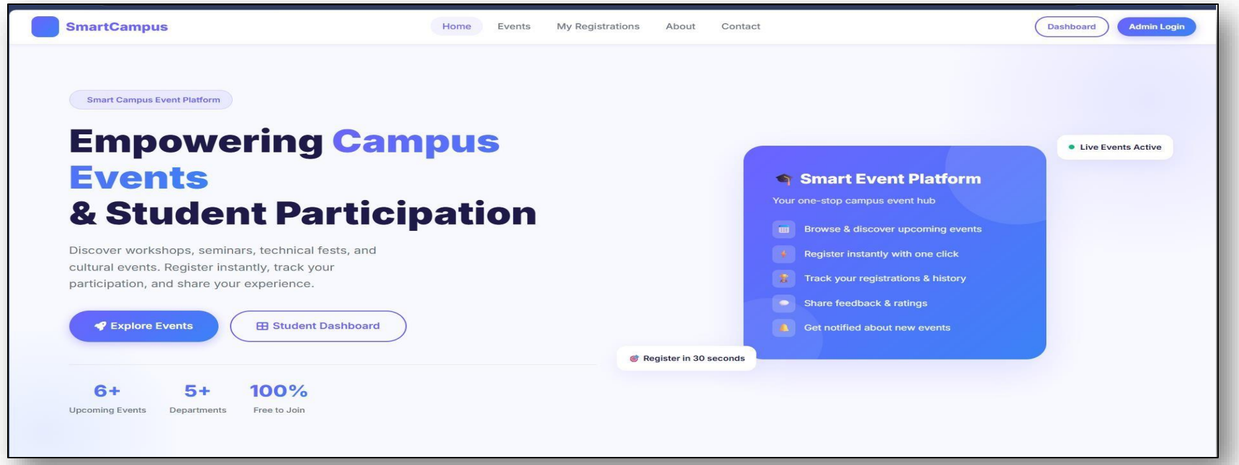
📝 Module 3: Registration & Feedback

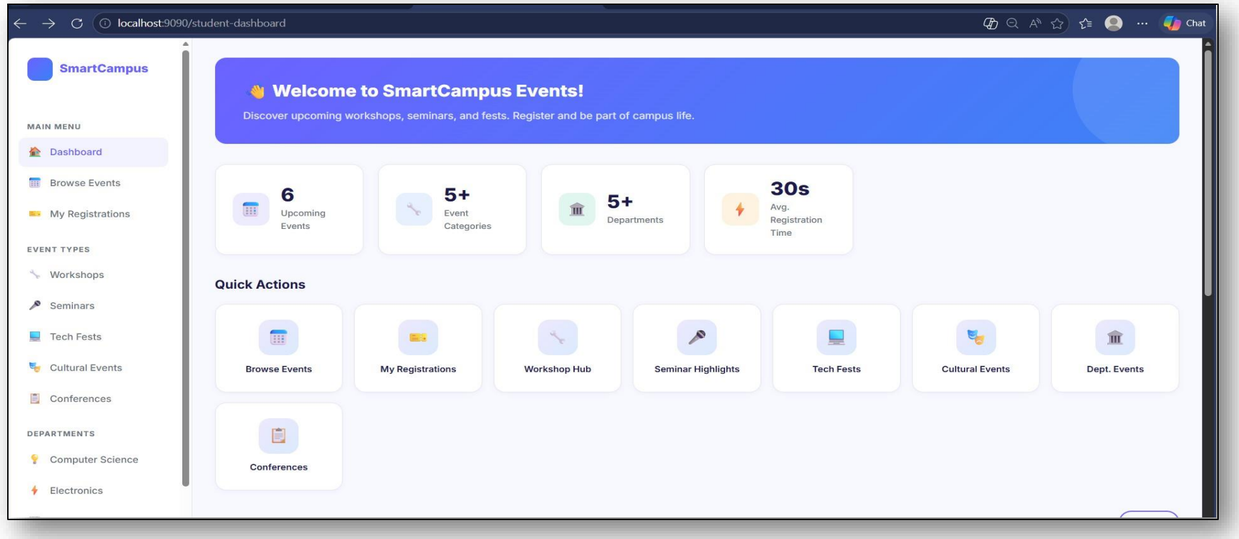
- Manages event registration and stores participant details.
- Allows users to submit feedback for events and improves user interaction.

🗄 Module 4: Database Management

- Handles storage and retrieval of users, events, registrations, and feedback using MySQL.
- Ensures data integrity and efficient data management.

IMPLEMENTATION SCREENSHOTS





Home Page – Event Dashboard

- The Home Page acts as the main interface where users can view all available campus events.
- Each event displays details such as event name, date, venue, and description.
- Users can easily browse events and select an event for registration.
- The event data is dynamically fetched from the MySQL database using Spring Boot.
- The search and filter feature allows users to find events based on department, date, or keywords.

IMPLEMENTATION SCREENSHOTS

LOGIN & SIGNUP - User Authentication Module

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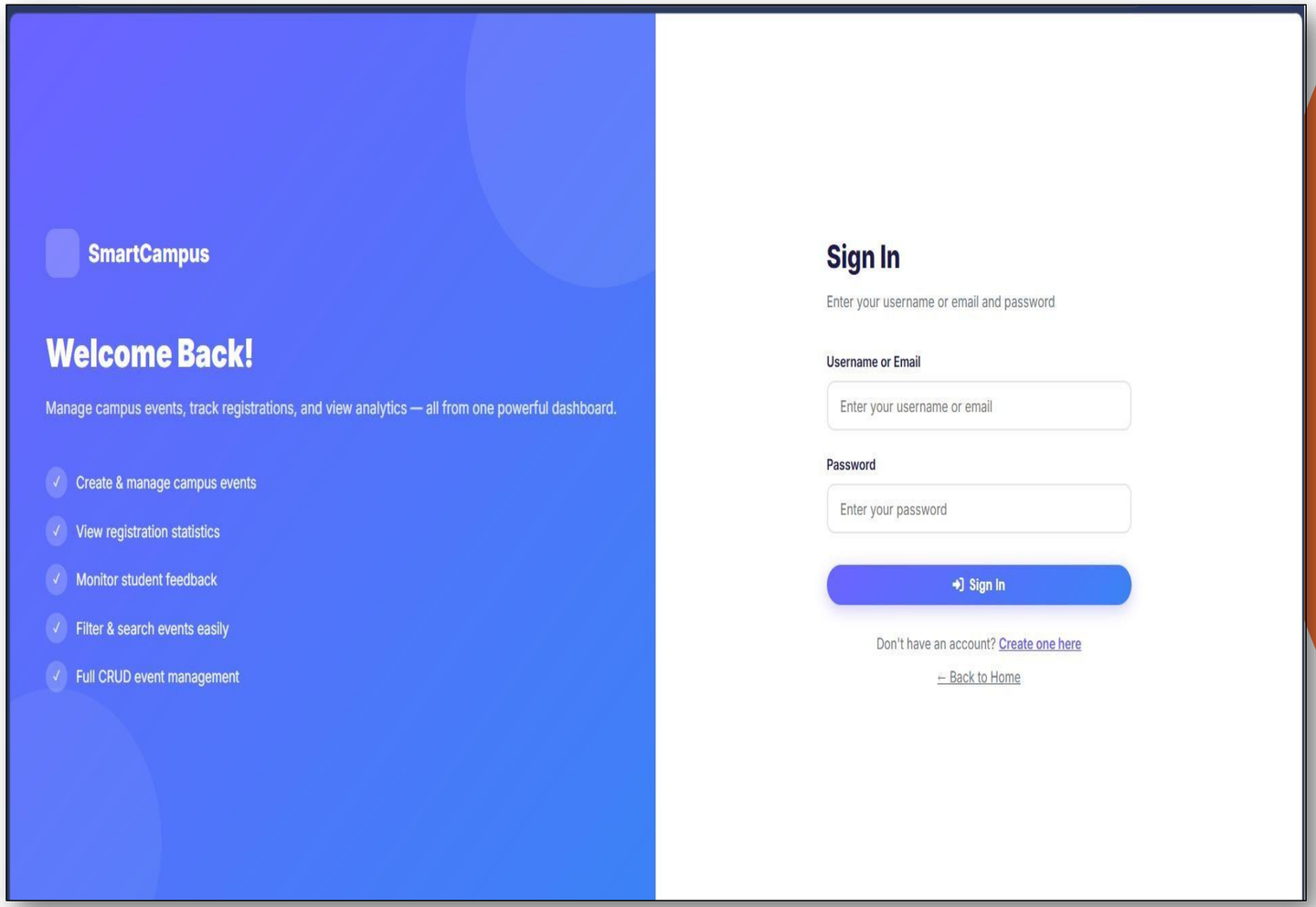
- The Login/Register page allows users to securely create accounts and log in.
- User credentials are validated using Spring Security authentication.
- Only authorized users (Student/Admin) can access the system features.
- Session management ensures secure and continuous user access.



IMPLEMENTATION SCREENSHOTS

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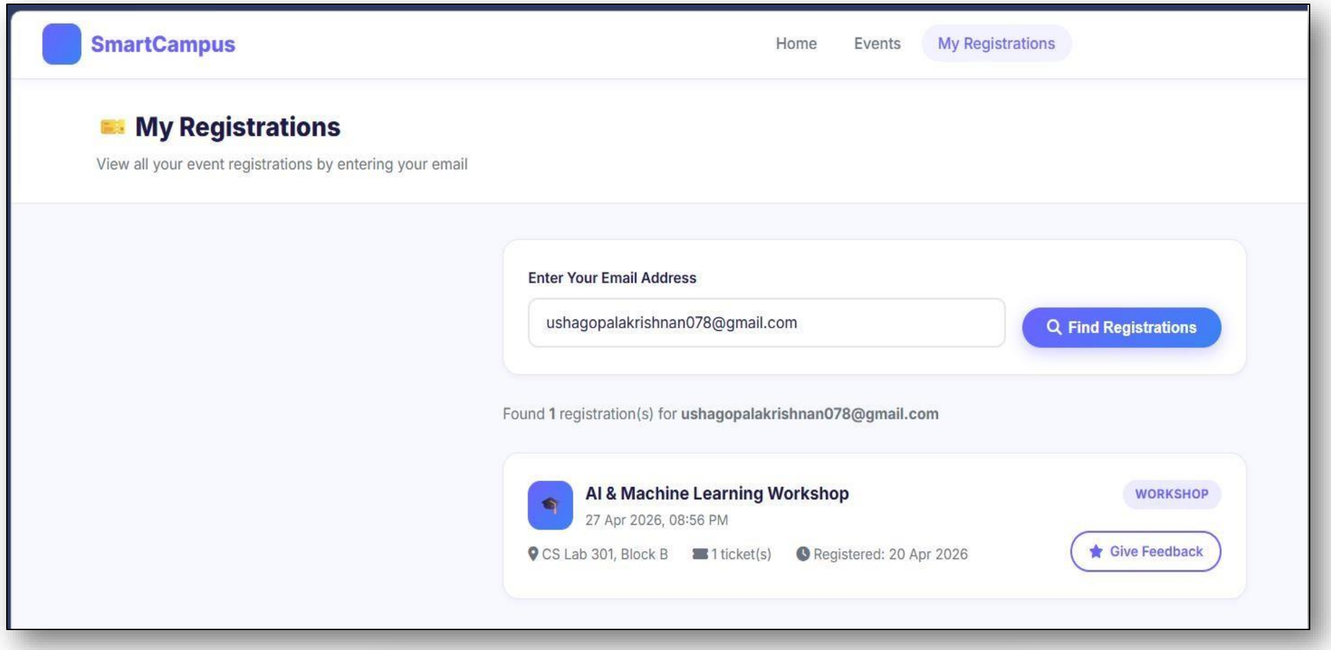
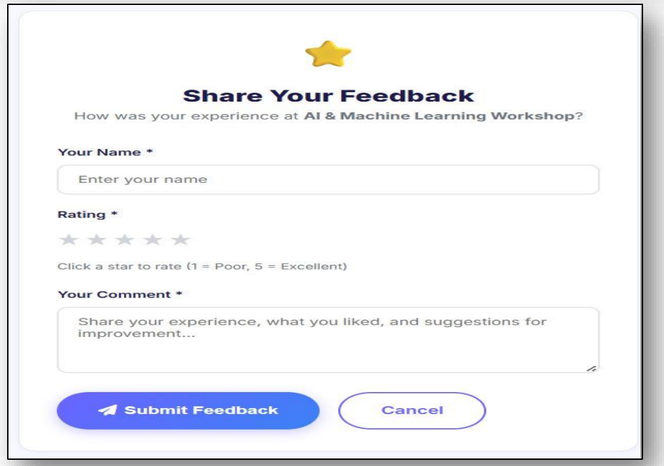
📝 Module 3 – Registration & Feedback

- ❖ Allows users to register for selected events by submitting required details.
- ❖ Stores registration data securely in the MySQL database.
- ❖ Enables users to submit feedback after participating in events.
- ❖ Ensures input validation to avoid incorrect or duplicate entries.
- ❖ Displays confirmation message after successful registration.





IMPLEMENTATION SCREENSHOTS

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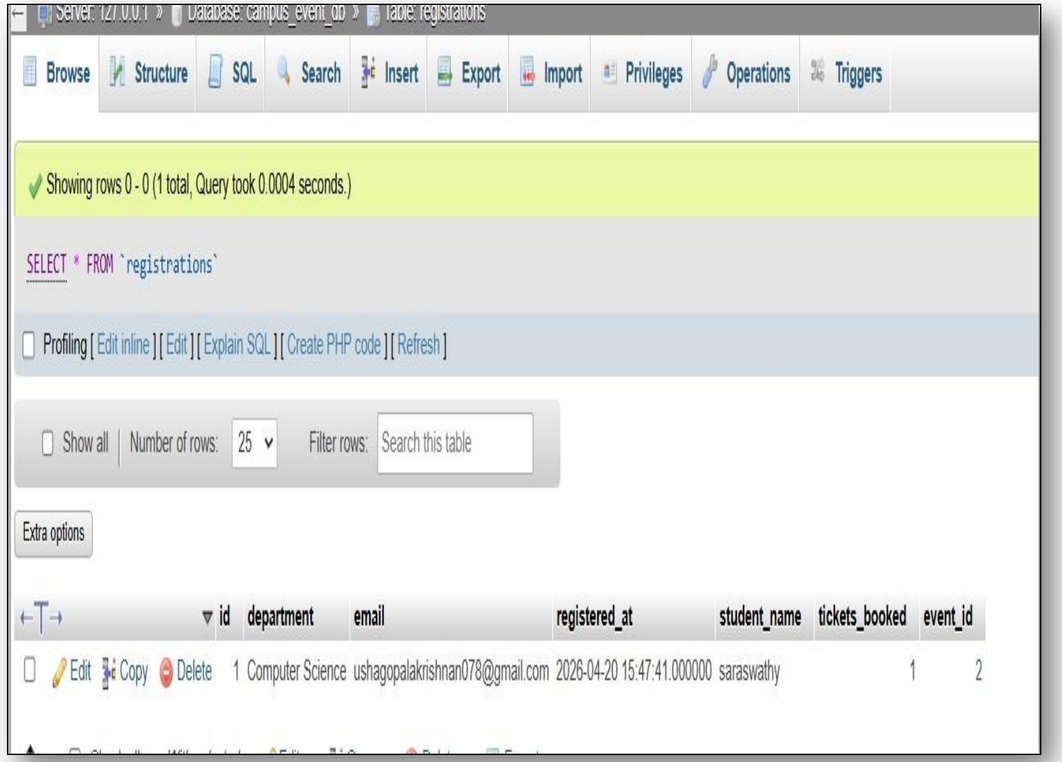

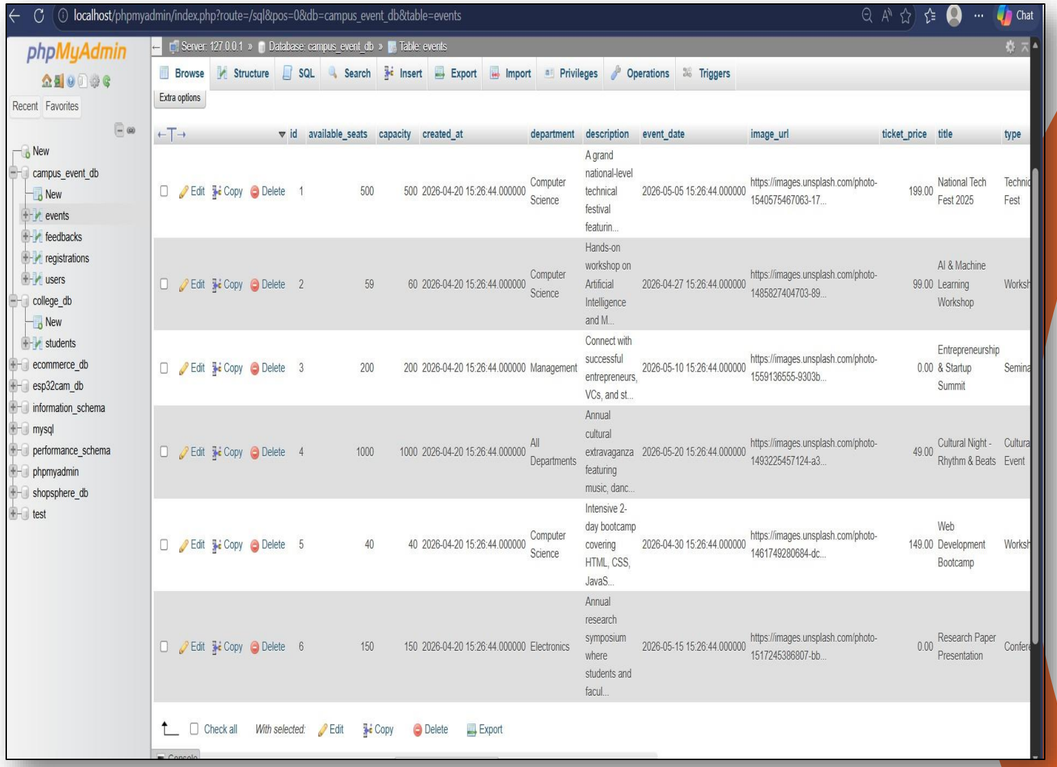

🗄 Module 4 – Database Management

- ❑ Handles storage and retrieval of user, event, registration, and feedback data using MySQL.
- ❑ Ensures data integrity and consistency through structured database design.
- ❑ Supports efficient CRUD operations (Create, Read, Update, Delete).
- ❑ Maintains secure data handling and prevents data duplication.
- ❑ Enables fast data access and real-time updates for the system.

ADVANTAGES & LIMITATIONS

✅ Advantages

- ✓ Provides a centralized platform for managing all campus events efficiently.
- ✓ Ensures secure authentication using Spring Security for user safety.
- ✓ Reduces manual work and errors in event registration and management.
- ✓ Enables real-time access to event details, registrations, and updates.
- ✓ Offers a user-friendly interface for easy browsing and registration.
- ✓ Supports data integrity and reliability using MySQL database.

⚠ Limitations

- ❑ Requires internet connection to access the system.
- ❑ Initial setup and configuration may require technical knowledge.
- ❑ Limited scalability if deployed only on local server (XAMPP).
- ❑ Security depends on proper configuration and user management practices.
- ❑ Does not include advanced features like notifications or mobile app support.

PHOTO

GPS Map Camera
New Vellanur, Tamil Nadu, India
53fw+hrv, New Vellanur, Tamil Nadu 600055, India
Lat 13.173841° Long 80.09704°
Tuesday, 21/04/2026 09:13 AM GMT +05:30
Google

Thank you